Snehitha Tadapaneni

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EXPERIENCE

Innomatics Research Labs — **Hyderabad, India**
Data Scientist Intern — **Jul 2023 – Jul 2024**

- Analyzed 100M+ records using Python (Pandas, NumPy), SQL, and Excel to identify demographic, performance, and recruitment trends.
- Automated data cleaning, validation, and reporting pipelines with SQL and Excel, reducing manual analysis effort by 30%.
- Built time-series forecasting models on operational delivery data and delivered Tableau dashboards to support data-driven decisions.
- Conducted exploratory data analysis (EDA) and statistical hypothesis testing (t-tests, chi-square) on 40,000+ records to evaluate business outcomes.
- Applied NLP techniques and SQL queries to analyze unstructured text data and improve compliance tracking and reporting accuracy.

SMIDR Lab — **GWU, Washington DC**
Graduate Research Assistant — **Aug 2025 – present**

- Analyzed human-robot interaction data using Python (Pandas, NumPy) and SQL to assess the impact of UI/UX decisions on user behavior and task outcomes.
- Collected, cleaned, and structured user interaction logs, building Python/SQL data pipelines for analytics and ML-ready workflows.
- Conducted EDA and pattern analysis with Pandas, Matplotlib, Seaborn, delivering insights through visual reports and dashboards (Tableau / Power BI????).
- Partnered with faculty to define KPIs and metrics, translating qualitative interaction data into actionable analytical insights.

Introduction to Engineering Computations(School of Engineering and Applied Sciences) — **GWU, Washington DC**
Graduate Teaching Assistant — **Jan 2026 – present**

- Led data-focused lab sessions for 150+ students covering Python, data cleaning, EDA, data visualization, and linear regression.
- Guided students through file-based data ingestion, transformation, and preprocessing, simulating real-world analytics pipelines.
- Debugged and optimized Python scripts during office hours, reinforcing statistical reasoning and analytical problem-solving.
- Evaluated data-driven projects, exams, and presentations, assessing model logic, data quality, and interpretation of results.
- Partnered with instructors to analyze student performance data and improve course materials and learning outcomes.

TECHNICAL SKILLS & CERTIFICATIONS

- **Programming & Scripting:** Python (Pandas, NumPy, SciPy, Scikit-Learn, Matplotlib, Seaborn), SQL, T-SQL, R, Databricks
- **Data Processing:** ETL, EDA, Data Cleaning, Data Transformation, Feature Engineering, Data Modelling, Statistical Analysis, Hypothesis testing, A/B Testing, Data Validation
- **Database Technologies:** Microsoft SQL Server, MySQL, PostgreSQL, MS Access, MongoDB
- **Data Visualization & Business Intelligence:** Tableau, Power BI, Excel (Advanced), Alteryx, Looker, DAX, VBA, KPI Dashboard Design, Data Storytelling, D3.js, Gephi, Google Analytics, ArcGIS, Big Query
- **Cloud & DevOps:** AWS, Microsoft Azure, Redshift, S3, EC2, Google Cloud Platform, Git, JIRA, CI/CD
- **Certifications:** AWS Certified Cloud Practitioner (Jul 2023), Tableau Desktop Specialist, Tableau, Google Data Analytics Professional Certificate, Google Business Analytics with PowerBI, Excel by SmipliLearn (Jun 2023), Foundations of Data Science by Google (Aug 2023), Python for Data Science and ML Bootcamp by Udemy (Nov 2023)

TECHNICAL AND RESEARCH PROJECTS

Environmental Insights from the NYC Street Tree Census & Air Quality Data **| Tableau | Excel | SQL**

- Analyzed NYC Tree Census & Community Air Survey data to study spatial links between tree health, biodiversity, and air quality.
- Designed an interactive Tableau dashboard visualizing tree density, species diversity, pollution, and stewardship across 59 districts.
- Performed regression analysis ($R^2 = 0.557$, $p < 0.0001$) revealing an inverse relationship between species diversity and PM2.5.

Analysis of crime rates on residential property in Washington DC in 2014-18 **| Python | SQL**

- Conducted an in-depth analysis of the relationship between neighborhood crime rates and residential property values using a dataset of 46,252 records and 39 attributes.
- Developed classification models (Random Forest, XGBoost) to segment neighborhoods by housing price tiers and income levels.
- Designed SQL queries to join, clean, and aggregate Open DC Data and Kaggle data across census tracts and longitudinal time dimensions.

Global Air Quality Data Analytics & Visualization **| Python | GCP | Looker Studio | Python**

- Built an end-to-end data analytics pipeline on GCP using Python and BigQuery to clean, transform, and analyze global air quality data.
- Performed SQL-based analysis to identify pollutant trends and regional AQI patterns across countries.
- Developed interactive Looker Studio dashboards to visualize AQI metrics and support data-driven environmental insights.

EDUCATION

The George Washington University — **Washington DC**
Master of Science, Data Science (GPA: 3.93/4.0) — **Aug 2024 - May 2026**

Vellore Institute of Technology (VIT) — **Vellore, India**
Bachelor of Technology, Computer Science and Engineering (GPA: 8.81/10.0) — **Aug 2020 - May 2024**